

CONTENTS

- 1 What we hold now
- 2 What it lets us do
- 3 The oracles
- 4 The price chain
- 5 Units, scales, grades
- 6 The law
- 7 The business
- 8 How a sale runs
- 9 What things cost
- 10 The ultimate
- 11 Questions for myself
- 12 Next ninety days
- 13 Glossary

Everything Lumber.bid

What the government price data now lets us do, how wood is actually priced from the stump to the shelf, the laws and costs around it, where this ends up, and the questions I would ask myself if I could.

2026-09-09 Session Walnut Prod project uuzqezohkqgsbbxyzvvv

The one rule that outranks everything here: we do not imply a market that does not exist yet. Zero lumber lots have ever closed. Every figure below is labelled either **MEASURED** — read from our own tables or a cited public document — or **RULE OF THUMB** — an industry order of magnitude you must verify locally before it goes in front of a customer.

Section 1

What we hold now

Three months ago the platform had one price table and it was empty. Today it holds four government sources in production, all re-read on their own schedules, all with the citation stored beside the number.

12,693

public timber sale results
with purchaser and price —
MI, WA, MN, MT

10,629

stumpage price rows,
Michigan 2009–2025, every
one with sales count and
range

21,447

source documents indexed
(notices, bid summaries,
prospectuses, reports)

44

catalogued government
sources across 22 states,
with probe status

All four counts read from prod on 2026-09-09. MEASURED

TABLE	WHAT ONE ROW IS	WHERE FROM	REFRESHED
gov_timber_sale_results	One sale: who bought it, what they paid, what it was appraised at, the outcome (sold / unsold / canceled)	MI DNR results (11,918), MN DNR notices (563), WA DNR results (193), MT DNRC bid summaries (6, FY26–27 only)	Weekly indexers, hourly extract + parse sweepers
public_timber_prices	One published average: species × product × unit × place × period, with sales count, volume, appraisal, min/max, and whether outliers were removed	MI DNR Average Stumpage Price Report, 17 annual files	Monthly, the 9th
gov_cut_and_sold	One region × species × product × fiscal quarter: volume and value sold and cut on National Forest land	USFS Cut & Sold reports (cumulative quarters — Q2 contains Q1)	Monthly
gov_source_documents	One PDF/CSV/XLSX the state published, with extracted text and a parsed-at stamp	The IIS trees and listing pages of the agencies above	Weekly
gov_sources	One agency URL we know about, with what it publishes and whether it still answers	Hand-catalogued 2026-08-30, 2026-09-08	On probe

Two readers exist for the outside world. timber_price_band(species, product, jurisdiction, months) returns low / median / high per unit with the source URLs, callable as anon. And the price table itself is readable by anon on purpose: it is public

record, and the estimator on `lumber.bid` is meant to read it. Nothing on `lumber.bid` calls either yet.

Section 2

What it lets us do

Each of these was impossible on 2026-08-23, when the price table was created empty because that session refused to type numbers from memory.

Cite a real price, with its evidence

"Basswood sawtimber in Michigan: \$77.60/MBF statewide in 2025 from 77 sales, range \$9–\$243; \$107.37 in 2024 from 86 sales." Every number links to the DNR spreadsheet it came from. That sentence is the difference between an appraiser and an opinion.

MEASURED

Show what competition is worth

On sold Minnesota tracts the median winning bid is **31% over appraisal**. That is the entire pitch for a sealed-bid marketplace, and it is a government number, not ours. The equivalent Michigan figure is computable from the same table the day we want it.

MEASURED

Name the buyers

12,693 sale results carry a `purchaser_name`. That is a list of every mill, logger and broker that bought public timber in four states, ranked by how much they spend, with the species they bought. It is the Phase B outreach list — from **timberbid.app**, never from this domain — and it did not exist a week ago.

MEASURED

Anchor the estimator to record instead of memory

`site/log-model.js` prices a log from a hand-built table. It can now show a band beside its answer — "Michigan public sales, statewide, last 24 months" — and the band is honest by construction because it cannot exist without a citation. This is the smallest next build with the largest honesty gain.

See a species over sixteen years

Michigan's series runs 2009–2025 at four aggregation levels. Basswood statewide sawtimber: \$129 in 2023, \$107 in 2024, \$78 in 2025. Red oak 2025: \$208 from 80 sales. That is a chart, a newsletter, and a reason for a landowner to come back. MEASURED

Read the federal floor by region

USFS FY2026 Q1: Grand Fir sawtimber in the Northern Region sold at \$180.63/MBF on 1,013 MBF. Federal timber sells cheap and thin; it is a floor, not a market. The 100 MBF volume floor in `usfs_stumpage_latest` exists so a 25 MBF sale does not print a price.

MEASURED

Section 3

The oracles — who publishes what wood sold for

"Reliable" here means: a government body, a public URL, a number tied to an actual sale or an actual sample of sales, and a method note. Everything else is somebody's opinion or somebody's product.

SOURCE	WHAT IT SAYS	PUBLIC?	STATUS WITH US
USFS Cut & Sold (national)	Sold and cut volume/value by region, species, product, fiscal quarter	Yes	INGESTING
Michigan DNR stumpage price report	Average sold price with count, volume, appraisal, range; 2009–	Yes	INGESTING
Michigan DNR sale results	Every state sale: purchaser, price, appraisal	Yes	INGESTING
Minnesota DNR notices (completed)	Per tract: purchaser, appraised total, bid total, outcome; 2016–	Yes	INGESTING
Montana DNRC bid summaries	Every bidder ranked, \$/ton, minimum bid, tons, MBF equivalent	Yes (FY26–27 reachable)	INGESTING
Washington DNR sale results	Auction results by sale	Yes	INGESTING
Minnesota DNR public stumpage price review	Annual statewide/regional price summary across public agencies	Yes	CATALOGUED
Wisconsin DNR forest products price reports	Stumpage and (historically) delivered prices by zone	Yes	CATALOGUED
Oregon Dept. of Forestry log price report	Pond value — delivered log prices by species and region	Yes	JS SHELL; NEEDS HEADLESS FETCH

Penn State / PA Timber Market Report	Quarterly stumpage and delivered prices from a buyer survey	Yes	JS SHELL
New York DEC stumpage price report	Semiannual stumpage by region and species, survey-based	Yes	JS SHELL
Vermont, New Hampshire, Maine, Connecticut reports	State/extension stumpage reports; NH via timber-tax filings, ME annual by county	Yes	JS SHELLS / PDFS; NEXT ROUND
NC State, Ohio State, other extension reports	Survey-based regional stumpage and delivered prices	Yes	CATALOGUED
Idaho Dept. of Lands sale results	Endowment sale results with bidders	Yes	CATALOGUED
TimberMart-South / TimberMart-North	The private benchmark for the South and Lake States	No — subscription	DO NOT INGEST
Hardwood Market Report, Random Lengths (Fastmarkets), Forest2Market	Lumber and log price services	No — subscription	DO NOT INGEST

Three things the catalogue taught. First, **recorded URLs rot**: both MN and MT links from 2026-08-30 were dead nine days later and had to be re-found. Second, **the extractor's text is not the page's text** — unpdf reversed one Montana file and not another, so every parser is label-anchored and order-independent. Third, **the survey reports (PA, NY, extension) are opinions of buyers averaged**; the sale-result reports (MI, MN, MT, WA) are transactions. We hold the transactions. That is the better side to be on.

The gap that matters most is in the Oregon row: **pond value** is a *delivered* log price, and every other oracle we hold is *stumpage*. Section 4 is why that distinction is the whole business.

The price chain — why stumpage is not a log price

A log has five prices depending on where it is standing, lying or stacked. Our oracle data is the first one. Lumber.bid sells the second, third and fourth. If we ever show a stumpage number as "what your log is worth" we will be wrong by a factor of two to five, in the seller's favour, and the honesty rule dies.

Stumpage Standing tree, on the stump, buyer does everything what we ingest	Delivered log At the mill gate or landing: stumpage + felling + skidding + loading + haul ≈ 2-4× stumpage	Green lumber Sawn, rough, wet. Log ÷ recovery + sawing cost ≈ 1.5-2.5× log	Kiln-dried lumber Dried to 6-8% MC, graded, often surfaced + \$0.15-0.50/BF, -7-10% volume	Retail / maker Per board, per slab, per project. Where live-edge lives 2-5× wholesale
--	---	--	--	---

Multipliers are **RULE OF THUMB**, US hardwood, 2020s; they vary wildly by species and grade. The first box is the only one we can prove today.

Two consequences the whole plan should stand on:

- **Logs are local; lumber travels.** A log truck earns its cost inside roughly 50-100 miles of a mill; beyond ~150 miles a log deal almost never pencils, which is why every mill has a "procurement radius". Kiln-dried, graded, bundled lumber ships across the country on a flatbed and is priced nationally. A national marketplace is therefore a **lumber** marketplace with a local log on-ramp — exactly the "bulk lumber is the industry focus" call, now with the reason underneath it. **RULE OF THUMB**
- **Urban and arborist logs are a different market with different rules.** Mills reject yard trees on sight (metal, short lengths, no history). Their path is a portable sawyer, a kiln, and a maker — the slab channel. That is the customer who opens `/estimate`. The stumpage oracle tells them nothing about their walnut; a delivered-log or lumber oracle would. We do not have one yet.

Worked chain, so the shape is felt

Red oak, Michigan, 2025: statewide sawtimber stumpage **\$208/MBF** from 80 sales.

MEASURED Add logging and haul at, say, \$150-250/MBF and the delivered grade log sits around \$400-500/MBF; a hardwood mill recovering ~60-65% and selling FAS/1C green

might see \$1,200–2,000/MBF lumber value from it; kiln-dried 4/4 FAS red oak at a retail yard runs a few dollars a board foot. **RULE OF THUMB** Every step past the first is a guess until we ingest a delivered-price series. That is the point of the exercise.

Section 5

Units, scales and grades — the vocabulary the data is written in

Board feet and the three log rules

A board foot is 12" × 12" × 1"; **MBF** is a thousand of them. A log's board-foot content is *estimated* by a scaling rule, and the rule you pick moves the price:

- **Doyle:** $(D - 4)^2 \times L \div 16$ (D small-end inches, L feet). Simple, legally common in the East and South, and it **undersizes small logs badly** — a 12" log scales at 32 BF Doyle versus ~50 International. Buyers like Doyle for a reason.
- **Scribner:** a diagram-based table; the Pacific Northwest and most federal sales use Scribner (often "Scribner Decimal C").
- **International 1/4":** the most accurate, used by many states and the USFS in the East. What a fair marketplace should quote, with the rule named on every lot.

Michigan's report is in MBF and cords; Montana sells by the **ton** and prints an MBF equivalent; the USFS reports both MBF and CCF (hundred cubic feet). Conversions are species- and log-size-dependent: a cord of hardwood pulp is ~2.5–2.8 green tons; a Doyle MBF of hardwood sawlogs is on the order of 8–10 tons. **RULE OF THUMB** Our tables never convert; they store the unit the agency used. Keep it that way.

Products in the agency vocabulary

Sawtimber (sawlogs, veneer logs at the top), **pulpwood**, **bolts** (short sawn products), **biomass**, **fuelwood**, poles and posts. Veneer logs are the prize — walnut, cherry, white oak, hard maple — and are priced per log, not per MBF, by the buyer standing in front of it. No public series captures veneer well.

Hardwood grades (NHLA)

GRADE	MIN BOARD	CLEAR-FACE YIELD	WHO BUYS IT
FAS (Firsts and Seconds)	6" × 8'	83⅓% in large cuttings	Furniture, millwork, export
F1F / Selects	4" × 6'	FAS one face, 1C the other	Furniture, mouldings
No. 1 Common	3" × 4'	66⅔%	Cabinets, flooring, "the workhorse"
No. 2A/2B Common	3" × 4'	50%	Flooring, pallets, crating
No. 3A/3B Common	3" × 4'	33⅓% / 25%	Pallets, industrial

Thickness is quoted in quarters: 4/4 is one inch rough, 8/4 two inches. Grades are assigned on the poorer face for most, the better face for FAS. Moisture matters as much as grade: green is above ~30% MC, air-dried lands at 12–20%, kiln-dried for interior use at 6–8%. A listing that says "kiln-dried" without a meter reading is a claim, not a fact — the seller wizard should ask for the number.

Softwood grades

Structural lumber is graded under WWPA/SPIB/NLGA rules — Select Structural, No. 1, No. 2, Stud — and stamped; appearance grades (C & Better, D) for boards. Grade stamps require a certified mill: a portable-sawyer's 2×10 cannot be sold as structural No. 2 and should never be listed as such.

Slabs and live edge

Priced per piece by species, width, length, figure, and dryness; a 3"-thick kiln-dried walnut slab 30" wide sells for many hundreds to a few thousand dollars, and the same slab green for a fraction. The photo is the product: end grain, full length, both faces, a tape in frame, and any metal or checking. That is why the wizard demands three photos before a lot goes live.

The law — what binds a sale, a harvest, and a marketplace

None of this is legal advice; it is the map of what to ask a lawyer about, with the statutes I am confident enough to name. Verify the current text before relying on any of it.

Federal

- **Lacey Act** (as amended 2008): trafficking in illegally taken plants, including timber and wood products, is a federal offence; a declaration is required on wood imports. Domestic effect: a marketplace should not be a fence for stolen wood, and a provenance line on every lot is cheap insurance.
- **OSHA 29 CFR 1910.266**, the logging operations standard: training, PPE, chainsaw and felling rules. Applies to anyone employing loggers; a marketplace does not, but a "we'll send a crew" promise would.
- **IRS §631(a)/(b)**: timber held over a year is capital-gain property; landowners need a **timber basis** and Form T; lump-sum sales of standing timber are reported on **Form 1099-S**; per-unit (pay-as-cut) contracts are §631(b) sales. Our platform will be asked whether it issues 1099s. Answer through Stripe's 1099-K rules for payouts and a lawyer.
- **Clean Water Act §404** silvicultural exemption — normal harvesting is exempt if BMPs are followed; skid trails through wetlands are not.
- **Endangered Species Act / CITES** touch a few species (some cedars, certain exports); not a factor for domestic hardwood lots.
- **Trucking**: 80,000 lb federal gross on interstates; log-truck tolerance permits are state matters. Michigan, Minnesota and Wisconsin impose spring "frost laws" — seasonal road weight limits that stop hauling for weeks. Prices and harvest windows bend around them.

State — the pattern, with named examples

INSTRUMENT	WHAT IT DOES	EXAMPLES
Forest Practice Act	Permit or notice before harvest, stream buffers, reforestation, licensed operators	Oregon FPA (15-day notification), Washington FPA (Forest Practices Application), California FPA (Timber Harvesting Plan; Licensed Timber Operator), Idaho FPA, Montana SMZ law, Maine Forest Practices Act notification, Massachusetts Forest Cutting Practices Act (licensed harvesters)
Intent to cut / yield tax	File before harvest; tax on stumpage value	New Hampshire RSA 79 (Intent to Cut, 10% timber yield tax); Vermont heavy-cut notice; Illinois 4% timber buyer fee
Timber buyer / logger licensing	Bond and license to buy timber or run a job	Indiana timber buyer license (IC 25-36.5, bonded); Illinois Timber Buyers Licensing Act; Kentucky Forest Conservation Act (Master Logger on site); Connecticut certified forest practitioners; Master Logger programs in ME, WI, MI and others
Timber trespass	Cutting the wrong tree costs multiples of its value	Treble damages: Washington RCW 64.12.030, Oregon ORS 105.810, New York RPAPL §861, Michigan MCL 600.2919; double/treble in Pennsylvania (42 Pa.C.S. §8311); California Civ. Code §3346
Property tax programs	Reduced assessment for managed forest, often with harvest rules and withdrawal penalties	Wisconsin Managed Forest Law, Michigan Qualified Forest / Commercial Forest, New York 480-a, Pennsylvania Clean & Green, Vermont Current Use
BMP regimes	Voluntary or mandatory water-quality practices; the thing a consulting forester actually enforces on the ground	Minnesota Forest Resources Council guidelines, Michigan SMZ practices, most Southern states voluntary with monitoring

Why it matters to a marketplace: the lot page should state which state the wood is in and, for standing timber, that the seller warrants title and the right to harvest, that the buyer handles notifications and permits, and which scaling rule and which party's scale governs

payment. For milled lumber none of the harvest law applies — another reason lumber is the easier product.

Marketplace law

- **UCC Article 2** governs the sale of goods; a lot description is an express warranty. "Approximately 2.4 MBF, International ¼", buyer may re-scale" is a description; "2,400 board feet" is a promise.
- **Marketplace-facilitator sales tax**: most states now make the platform collect tax on third-party sales above a threshold. Stripe Tax or equivalent before the first out-of-state lumber sale, not after.
- **Payout reporting**: Stripe Connect handles 1099-K for sellers above the federal threshold; state thresholds differ.
- **Escrow and deposits** are already single-source in timberbid-v1 (`manage-job-deposit`). Do not build a second money path here.

The business — who is in it and where we stand

ACTOR	WHAT THEY WANT	HOW THEY GET PAID	LUMBER.BID RELATIONSHIP
Family forest owner (~10 million in the US, ~1/3 of forestland)	A fair price once a decade; not to be cheated	Lump sum or pay-as-cut	Seller of standing timber (later); reader of the oracle (now)
Consulting forester	Marks, cruises, runs the sealed bid, supervises	~8–12% of sale value or per-acre RULE OF THUMB	The natural first power user of a sealed-bid rail; also a competitor to it
Logger / harvesting contractor	Steady stumpage within haul range; roads that hold	\$/MBF or \$/ton cut-and- haul; or buys stumpage and sells logs	Buyer of stumpage; seller of logs
Procurement forester (mill- employed)	Fill the log yard at the lowest delivered cost	Salary	The buyer the purchaser list names. The Phase B call.
Sawmill (large band; small circle; portable)	Logs at 60–75% of lumber revenue RULE OF THUMB	Lumber, chips, sawdust, bark	Buyer of logs; seller of bulk lumber
Arborist / tree service	To not pay to dump a walnut trunk	Removal fee; log is a by- product	Seller of urban logs — the /estimate visitor
Portable sawyer, kiln operator	Work by the hour or board foot	\$75–125/hr or \$0.35– 0.75/BF sawing; \$0.15– 0.50/BF drying RULE OF THUMB	Service partner on the slab channel; seller of slabs
Woodworker / maker	Specific boards, dry, photographed, shipped	Sells furniture	Buyer of slabs and small bulk lumber — the buyer we can actually reach online
Wholesaler / distribution yard	Truckloads of graded KD lumber	Margin	Buyer and seller of bulk lumber packs

How we earn, stated as copy, computed in one place

A 2.5% seller fee at settlement and a tiered buyer fee, both computed only in `utils/fees.ts` / `_shared/fees.ts`. The arithmetic that should sober every roadmap conversation: fifty \$3,000 lots in a year is \$150,000 of volume and about **\$3,750** of seller fee. The buyer tier adds some. This is a business only at volume, or at higher ticket (bulk packs, standing timber), or with a second revenue line (the oracle as a product for foresters, lenders, insurers, appraisers). Pick on purpose.

Where the moat is

Not the auction — anyone can build a form. The moat is the **ingest layer**: 21,447 documents indexed, parsers that survive unpdf, run rows that say what broke, and a table that cannot hold a price without its citation. Private services sell this; we can give it away, rank for it, and sit one click from the seller who just learned what their timber is worth.

Section 8

How a sale actually runs

A standing timber sale (the state pattern, which private sales copy)

1. **Cruise** — a forester samples the stand, estimates volume by species and product, and sets an appraisal.
2. **Prospectus / notice of sale** — species, volumes, scaling rule, minimum bid, bond, contract length (1–3 years), road and season restrictions, showing date.
3. **Showing** — bidders walk it. Nobody serious bids blind.
4. **Bid opening** — sealed bids read aloud, or an oral auction after sealed qualifying bids. Minnesota records every bidder; Montana ranks them.
5. **Award and bond** — performance bond posted; a percentage down.
6. **Harvest** — inside the window, around frost laws, fire season, nesting restrictions.
7. **Scale and pay** — lump-sum: paid up front; pay-as-cut: scaled at the mill and paid per unit, which is why "whose scale governs" is the clause that matters.
8. **Closeout** — roads, slash, BMP inspection, bond released.

Our sealed rail already mirrors steps 2, 4 and 5 (`auction_mode='sealed'`, deposits, anti-snip). What it lacks is step 3, the showing — which for a log or lumber lot is the photo set, and for standing timber is a real visit we cannot replace.

A log or slab sale (the lumber.bid fixed-price path today)

1. Seller photographs (end grain, full length, defects, tape), enters species, dimensions, quantity in MBF or pieces, grade if known, moisture if measured, and a price per MBF or per piece. The wizard asks one thing at a time because that is the only way people finish.
2. The lot goes active with three photos or stays a draft. A lead becomes a listing through a single-use claim link.
3. Buyer pays through the existing checkout; funds are held; seller ships or buyer collects; settlement releases and the 2.5% comes off.
4. Disputes hinge on the description: scale rule, moisture, count. Write descriptions as ranges and methods, never as bare numbers.

Practices that decide whether wood is worth anything

- **Buck for value:** a 16' log with a defect at 9' is worth less than a 8' clear log plus a 8' tie log. Log length + 4–6" trim allowance; mills publish spec sheets (minimum small-end diameter, allowed sweep, lengths).
- **Metal:** urban logs carry nails, fence wire, hooks. One strike ruins a blade (\$30–\$300) or a band; commercial mills refuse yard trees for this alone. A metal-detector pass is the cheapest thing a seller can do and should be a listing field ("scanned: yes/no").
- **End-seal within days** of felling (wax emulsion) or lose the last foot of every log to checking.
- **Sticker-stack** off the ground, in the shade, air-dried a year per inch of thickness as a rule, then kiln to 6–8% for interior work. **RULE OF THUMB**
- **Sort and grade at the landing:** veneer, grade sawlog, tie/pallet log, pulp. The first sort decides the price more than the auction does.

Section 9

What things cost

All **RULE OF THUMB**, US, 2020s, and wide because the country is wide. Use them to sanity-check a listing, never to price one.

ITEM	TYPICAL	NOTES
Consulting forester, sale administration	8–12% of gross, or \$/acre	Pays for itself on competitive sales; the MN 31% premium is the argument
Cut-skid-load, hardwood, conventional	\$80–200/MBF	Steep ground, small tracts and short logs push it up; cut-to-length mechanised crews price by the ton
Log haul	\$0.15–0.30 per ton-mile; \$40–120/MBF for typical hauls	This line sets the procurement radius
Road building / stream crossing	\$5,000–50,000+ per mile; a temporary bridge \$2,000–10,000	Often the difference between a sale and no sale
Tree removal with crane (urban)	\$1,500–6,000 per large tree	The log is a by-product; the arborist wants it gone
Portable sawmill	\$75–125/hr or \$0.35–0.75/BF; blades \$30–\$40 each	Metal strike = blade + time
Kiln drying	\$0.15–0.50/BF; 7–10% shrinkage; 1–8 weeks by species and thickness	Slabs and 8/4+ take months in a dehumidification kiln
Surfacing / straight-line rip	\$0.10–0.30/BF	
LTL / flatbed freight, lumber	\$150–600 per pallet regionally; a truckload (~20 MBF KD hardwood) \$1,500–4,000 cross-country	Lumber travels; logs do not
Logging equipment	Feller-buncher \$300–500k, skidder \$200–400k, forwarder \$250–450k, loader \$150–300k	Why loggers need steady stumpage within radius
Workers' compensation, logging	Among the highest class rates in the US; can exceed 20–40% of payroll in some states	Why "we'll send a crew" is never our promise

Mill log cost share	60–75% of lumber revenue	The mill's margin lives in recovery, chips and grade-out
Platform: payments	Stripe ~2.9% + 30¢ per card charge; Connect payout fees; disputes \$15 each	On a \$3,000 lot the card fee alone is ~\$87 — more than the 2.5% seller fee (\$75)
Platform: infrastructure	Supabase, Netlify, Resend, domain — low tens of dollars a month at this scale	Not the constraint

The row to stare at is the payments one: on small lots the card processor earns more than we do. Either lots are bigger, buyers pay the fee, ACH is the rail for bulk, or the fee unit changes. That decision belongs in the fee module and nowhere else.

Section 10

The ultimate — what Lumber.bid is when it is finished

The public record of what wood actually sold for, and the place you sell into it.

Three layers, each honest on its own, each feeding the next:

1. **The oracle.** Every public timber and lumber price in the country, cited, searchable, charted, free. Stumpage from the states and the USFS (have), delivered log prices (Oregon pond values, Wisconsin, PA, NY — next), and eventually our own closed lots as a published series. Foresters, lenders, appraisers, insurers and landowners use it because there is no free alternative and the private ones cost thousands a year. This is the top of the funnel and the SEO surface, and it is true whether or not a lot has ever closed.
2. **The appraisal.** The estimator, anchored to the oracle: a photo becomes species, diameter, length and condition; the band beside it is government evidence with a link. It never prices; it shows what similar wood sold for and lets the seller decide.
3. **The market.** Fixed-price listings for logs, slabs and bulk lumber in NHLA grades priced per MBF (live), then sealed-bid lots with deposits and escrow (rails live, no lots yet), then standing timber administered with a forester. Fees stay single-source. The purchaser list seeds the buyer side from timberbid.app.

What "won" looks like, in order

- One real listing from a stranger, with three photos. (Not yet.)
- One lot closed with competing bids. That day the leaderboard, the sitemap entry, the homepage card and the og record card all come back together — the four demotions lift as one.
- The oracle page ranks for "[species] stumpage price [state]" queries and a landowner arrives who has never heard of us.
- A named mill from the purchaser list buys a bulk pack.
- A closed-lot price series we publish is cited by someone else.

Non-goals, so they stay non-goals

- No logging crews, no hauling, no kilns of our own. We are a record and a market, not an operator.
- No private price data, ever. If it has a subscription, it is not ours to republish.
- No second money path, no second email path, no service-role key on a static site.
- No implied liquidity. The copy says "no auction is open yet" until it is false.

Section 11

Questions I wish I could ask myself

Asked plainly, with the honest answer and what it would take to know.

Is stumpage even the right oracle for the person who opens /estimate?

No. They have a felled walnut in a driveway, and stumpage is the price of a standing tree the buyer must go and cut. The number that matches their situation is a delivered-log or green-lumber price. We hold neither. Oregon's pond values are the first delivered series that is public; get it, then look for Lake States delivered prices in the Wisconsin reports.

Who, by name, is the first buyer?

It is in the table. `select purchaser_name, sum(sold_value) from gov_timber_sale_results group by 1 order by 2 desc` is the list. Rank by spend, filter to species we can supply, find the procurement forester, and call from timberbid.app. Until one of them says yes or no, "buyer side" is a phrase, not a plan.

Why did six people open the estimator and none finish?

We do not know, and we could. The beacons that would say (photo added, estimate requested, returned, rejected) have never fired, which means the drop is before the first photo. On a phone that is usually the camera permission or the upload size. Watch one real person try it. The scheduled analytics check on the 12th will say whether the problem is traffic or conversion; my bet is both.

Are we a logs business or a lumber business?

Lumber, with a local log on-ramp. Logs cannot leave a 100-mile circle economically; kiln-dried lumber is priced nationally and ships on a pallet. Every product decision should favour the thing that can be sold to a stranger three states away.

Does the fee even work?

Not on small lots. The card processor takes ~\$87 on a \$3,000 lot; we take \$75. The business is bulk packs and standing timber (ticket) or foresters running many sales (volume) or the oracle as a paid product for professionals (second line). Decide, then let `fees.ts` reflect it.

Is the sealed-bid auction a feature anybody asked for?

Nobody has, yet. Consulting foresters already run sealed bids by email and know their bidders. What they lack is reach beyond their bidder list and a clean deposit/escrow rail. Sell them those two things; do not sell them "an auction".

What is our liability when a lot's board-foot count is wrong?

Whatever the description promised. A count is an express warranty under UCC Article 2. Descriptions must carry the scaling rule, the word "approximately", and a buyer re-scale clause; the wizard should write that sentence for the seller.

What happens when Michigan changes its file format?

Today: a skip counter climbs, or a run row goes red, and nobody is told. The sources rotted once already in nine days. Add a "stale or shrinking source" alert on `gov_award_ingest_runs` — the same discipline the sawlog alert learned on 2026-09-05: a replay discriminator and a burst valve from day one.

Is a 24-month band window honest?

Borderline. Basswood went \$129 → \$107 → \$78 in three years. A 24-month median hides a falling market. Show the last two annual points as points, not a blended median, and let the reader see the slope.

Should the oracle be an open dataset?

Yes. It is public record with our citations attached. Publishing it (CSV per state, a stable URL, a method note) is the cheapest trust and SEO we will ever buy, and it costs nothing we sell.

Who eats the blade when there is metal?

The sawyer, unless the listing says otherwise. Make "metal-scanned" a field, show it on the card, and let the price carry the risk instead of the dispute queue.

Can the appraiser be wrong in a way that costs someone real money?

Yes, in both directions. A high band makes a seller refuse a fair offer; a low one makes them give walnut away. The defence is the same as the honesty rule: show sales count and range beside every number, never a single figure, and never a stumpage number for a log on the ground.

What is the smallest lot worth a sealed bid?

Unknown; probably a truckload. Below that, fixed price and a buy button. Sealed bidding earns its friction when several buyers would bid and the spread matters — roughly the point where the MN 31% premium exceeds the buyer's cost of looking.

What would make me delete this whole vertical?

Twelve more months of zero listings while the tree vertical is also starved. The canon says stage-3 verticals wait for a full funnel. The oracle should survive regardless — it costs nothing and helps timber.bid — but the market layer is a bet that a stranger will list. Set the date now.

Section 12

Next ninety days, if I were choosing

1. **Put the band on the estimator page**, statewide level only, with the citation and the sales count. One RPC call, one paragraph of copy, no new capability. The first honest price the site has ever shown.
2. **Ingest one delivered-price series** (Oregon pond values first; it needs a headless fetch) so the estimator can show a number that matches a log on the ground.
3. **Publish the oracle**: a /prices page per state and species, static, from the same RPC, plus a CSV. This is the SEO surface and it is true today.
4. **Run the purchaser list** and make ten calls from timberbid.app. Record every answer in dev_log.
5. **Add the stale-source alert** so the ingest layer cannot rot silently.
6. **Watch one real person use /estimate** on a phone. Fix what they hit.

Nothing on that list adds a capability the audit said was missing; every item makes something we already built visible, honest, or watched.

Section 13

Glossary

Stumpage

The price of standing timber, on the stump; the buyer fells, skids and hauls. What all our state data is.

Delivered price / pond value

The price of a log at the mill gate (or in the pond). Stumpage plus logging plus haul.

BF, MBF, CCF, cord, cunit

Board foot; thousand board feet; hundred cubic feet; 128 stacked cubic feet; 100 solid cubic feet.

Doyle, Scribner, International ¼"

The three log scaling rules. Doyle undersizes small logs; International is most accurate; Scribner rules the West.

Overrun / recovery

Lumber sawn versus log scale. Doyle logs "overrun"; the mill's margin lives here.

Cruise, prospectus, showing

The volume estimate, the sale document, the site visit before a bid.

Lump-sum vs pay-as-cut

Paid up front on the estimate versus paid per unit as scaled at the mill.

Mill tally / woods scale

Whose measurement governs payment. The clause that matters.

NHLA grades, 4/4, KD, MC

Hardwood lumber grades; thickness in quarters of an inch; kiln-dried; moisture content.

FAS, 1C, 2AC

Firsts and Seconds; No. 1 Common; No. 2A Common.

Veneer log, grade log, tie log, pallet log

Descending sorts of a sawlog; the first sort decides the price.

Slab, flitch, cant, live edge

A thick full-width board with bark edges; a log sawn through-and-through; a squared log; the natural edge left on.

SMZ / RMZ, BMP

Streamside / riparian management zone; best management practices for water quality.

Frost laws

Seasonal road weight limits in the Lake States that stop hauling in spring.

Trim allowance, sweep

Extra inches cut beyond nominal length; curvature along a log.

Cut & Sold

The USFS quarterly report of timber sold and cut on National Forests; quarters are cumulative.

Exclusion row

In Michigan's 2009–2016 reports, the outlier sales removed from an average and reported beside it. Not an alternate average.

Sources for measured figures: `public_timber_prices`, `gov_timber_sale_results`, `gov_cut_and_sold`, `gov_source_documents`, `gov_sources` in prod as of 2026-09-09; migrations 20260907041801, 20260908120000, 20260908130000, 20260908150000, 20260909040000; PRs #1087, #1095, #1100. Rule-of-thumb figures are industry orders of magnitude, US, 2020s, and must be verified locally before use with a customer. Legal references are pointers for counsel, not advice.